RISHI MADHUR KANAJAM

AI & Data Engineer
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SUMMARY

AI and data engineer who builds production-minded data pipelines and LLM systems with evaluation, validation and governance built in. Hands-on across Python ETL and data integration (SAP, Workday, Azure data services at St Vincent de Paul Society NSW), hybrid RAG with LLM-as-judge evaluation, agentic workflows with tool calling and structured outputs, and FastAPI services on AWS, Azure and GCP. Built a rules-driven payments recovery engine on Australian direct-debit data and maintain Krama Core, an open-source health-data compliance SDK on PyPI. Master of IT (Data Analytics & Management), University of Sydney; AWS Certified AI Practitioner. Full Australian working rights.

TECHNICAL SKILLS

Data Engineering: SQL, Python ETL/ELT, Pandas, NumPy, Spark, Airflow, Snowflake, data quality validation, data mapping and lineage documentation, PostgreSQL, MongoDB Atlas, SQLite, Redis
AI & GenAI: RAG (hybrid BM25 + dense, Reciprocal Rank Fusion, cross-encoder re-ranking), LLM agents and agentic workflows, LangChain, LangGraph, Semantic Kernel, prompt engineering, tool calling, structured outputs, grounding and output validation, LLM-as-judge evaluation; Anthropic, OpenAI, Gemini and Groq APIs; vector search (MongoDB Atlas, ChromaDB)
Machine Learning: PyTorch, Hugging Face, scikit-learn (Isolation Forest, One-Class SVM), YOLOv8, TensorRT, INT8 quantisation, pruning, inference profiling
Backend & APIs: Python (primary), FastAPI, Flask, Pydantic, Alembic, REST APIs, WebSockets, idempotent webhooks, Node.js
Cloud & DevOps: AWS (Certified AI Practitioner), Azure (AI Language, data services), GCP (Cloud Run, Vertex AI), Docker, GitHub Actions CI/CD, pytest, ruff, bandit, pip-audit
Governance & Security: Compliance rule engines, encryption (ECDH X25519 + AES-GCM), audit-ready decision logging, FHIR R4 health-data standards

PROFESSIONAL EXPERIENCE

Academic Tutor, Data Structures & Algorithms **01/2024 - Present**
University of Sydney

- Teach trees, priority queues, graph algorithms, dynamic programming and complexity analysis to 50+ undergraduate and postgraduate students per semester.
- Improved lab materials and test cases; built an interactive React platform that visualises data structures step by step.

Systems Engineer - Data Team (Part-time) **04/2025 - 04/2026**
St Vincent de Paul Society NSW (Vinnies)

- Built Python scripts to automate data extraction, cleansing and validation, reducing manual effort across the team's ETL processes.
- Supported data integration workflows across SAP, Workday and Azure data services: mapped source-to-target fields, validated outputs and troubleshot pipeline failures.
- Documented data flows and transformation logic so pipelines could be maintained and audited by the wider team.

Computer Vision Engineer (Industry Capstone, Team of 4) **08/2024 - 12/2024**
Ocius Technology

- Owned model optimisation in a four-person team building a real-time YOLOv8 maritime object-detection pipeline for Ocius's autonomous Bluebottle vessels.
- Applied INT8 quantisation, pruning, TensorRT, batching and inference profiling, delivering the optimised model behind a ~9× team-wide throughput gain on NVIDIA Jetson edge hardware; containerised the pipeline with Docker.

Frontend Developer **04/2023 - 07/2023**
HRVITE Services Pvt Ltd

- Built backend REST APIs and data-driven Angular components for an HR SaaS platform handling nested HR datasets; UI re-engineering contributed to a ~15% increase in user retention.
- Wrote unit tests and took part in code reviews to reduce regressions.

Software Engineer Trainee **07/2022 - 01/2023**
Revidd

- Moved from a two-month traineeship into full engineering responsibilities at an early-stage startup; built a new platform frontend and worked on CI/CD pipelines for a microservices architecture.

KEY PROJECTS

PDF-Rag - Hybrid Retrieval with Evaluation | link

- Combined BM25 and dense retrieval with Reciprocal Rank Fusion and cross-encoder re-ranking, plus sentence-aware chunking and conversational memory.
- Built an LLM-as-judge evaluation harness scoring faithfulness, answer relevancy and context quality, so retrieval changes are measured rather than eyeballed.

Pinch Recovery Engine - Failed-Payment Recovery **|** Pinch 50K Hackathon | link

- Classifies Australian direct-debit dishonour codes and applies a recovery strategy per failure class: never retries hard failures and schedules retries only on Australian business days.
- Logs human-readable reasoning for every decision; FastAPI, PostgreSQL, Alembic, Docker, idempotent webhooks, runs in mock mode or against the live Pinch sandbox.

Krama Core - Open-Source Healthcare Infrastructure | link | PyPI

- Python SDK for India's ABDM health network: consent and data-exchange flows, FHIR R4 bundle builders, ECDH X25519 + AES-GCM encryption, gateway retries and circuit breakers.
- Compliance engine returns blockers, not warnings, with country rule packs; v1.0.0-alpha.3 on PyPI, Apache 2.0, 128 tests, CI on Python 3.10–3.12.

PathFinder - AI Triage & Referral Platform **|** NGM Group Hackathon, Idea Disruptor Award (Team) | link

- Intake chatbot, referral portal and staff dashboard for a regional suicide-prevention network using FastAPI, Azure AI Language and React.
- Rule-based risk engine routes referrals across 14 programs with a five-layer escalation protocol and 35+ documented edge cases.

DeceptionArena - LLM Behaviour Benchmark | In progress, solo

- Benchmark infrastructure running five social deduction games as state machines behind one engine interface, with structured decision traces and 15 models across 5 providers wired in; 1,000/1,000 randomised stress tests pass
- Tournament ELO system, React + Pixi.js visualiser, structured evaluation outputs. The goal is a reproducible evaluation that captures behavioural differences between models, not just accuracy, but how models fail and why. Targeting arXiv publication.

MedVault - Multimodal Clinical Decision Support **|** link

- Built a solo multimodal agent that ingests clinical PDFs, images, and free-text notes through Gemini 1.5 Pro, with embedding-based retrieval grounding to reduce fabricated citations.
- Designed a structured tool-use protocol so the model produces auditable outputs, citations, retrieval traces, and confidence flags, patterns directly applicable to customer-facing AI in regulated domains.

EDUCATION

Master of IT & IT Management, Data Analytics & Management **07/2023 - 07/2025**
University of Sydney

Bachelor of Technology (Honors), Computer Science and Engineering **08/2019 - 04/2023**
GMR Institute of Technology

CERTIFICATIONS

- AWS Certified AI Practitioner - AWS (1bedf4d1017b450ebbbd31bcb452868f)
- McKinsey Forward Program (Apr–Jun 2026) - (business problem-solving, communication, leadership)